

Using Python for Data Mining

This article presents a few examples on the use of the Python programming language in the field of data mining. The first section is mainly dedicated to the use of GNU Emacs and the other sections to two widely used techniques—hierarchical cluster analysis and principal component analysis.

This article is introductory because some topics such as varimax, oblimin, etc, are not included here and will be discussed in the future. The complete code is too long for a printed article, but is freely available at <https://github.com/astonfe/iris>.

The toolbox used in this article is dependent on WinPython 3.4.4.2 and GNU Emacs 24.5 on Windows. My Emacs configuration for the Python language is very simple. The following lines are added to the *dot emacs* file:

```
(setq python-indent-guess-indent-offset nil)
(org-babel-do-load-languages
 'org-babel-load-languages
 '((python . t)))
(add-to-list 'exec-path "C:\\WinPython-
32bit-3.4.4.2\\python-3.4.4")
(global-set-key (kbd "<f8>") (kbd "C-u C-c
C-c"))
(setenv "PYTHONIOENCODING" "utf-8")
(setenv "LANG" "en_US.UTF-8")
```

The first line is useful to avoid the warning message: 'Can't guess python-indent-offset, using defaults: 4' from Emacs. The next three lines are to use Python in the org-mode, and the last four lines are to use Emacs as an IDE. In the following org file, text, code, figures and a table are present at the same time. This is not very different from a *Jupyter Notebook*. Each code section can be evaluated with C-c C-c. The export of the whole file as HTML (C-c C-c e h h) produces the output shown in Figure 1.

```
#+title: PAH method GCMS
#+options: toc:nil
#+options: num:nil
#+options: html-postamble:nil
```

```
<...some-text-here...>
#+begin_src python :var filename="method.png" :results file
:exports results
<...some-python-code-here...>
#+end_src
#+results:
[[file:method.png]]
#+begin_src python :var filename="chromatogram.png" :results file
:exports results
<...some-python-code-here...>
#+end_src
#+results:
[[file:chromatogram.png]]
#+attr_html: :frame border :border 1 :class center
<...a-table-here...>
```

To do this, it is necessary that Python is recognised by

